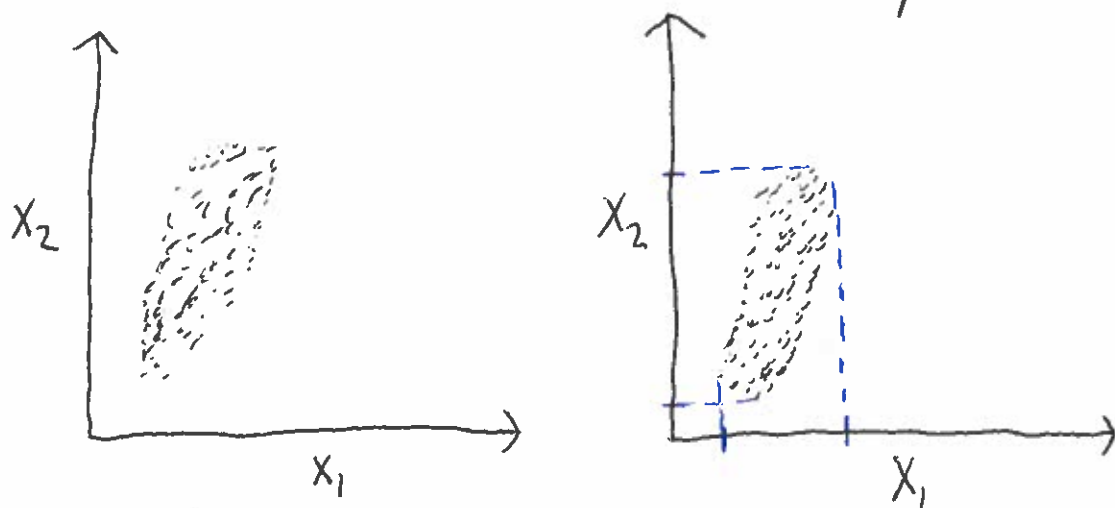


Visual Explanation: Principal Components Analysis

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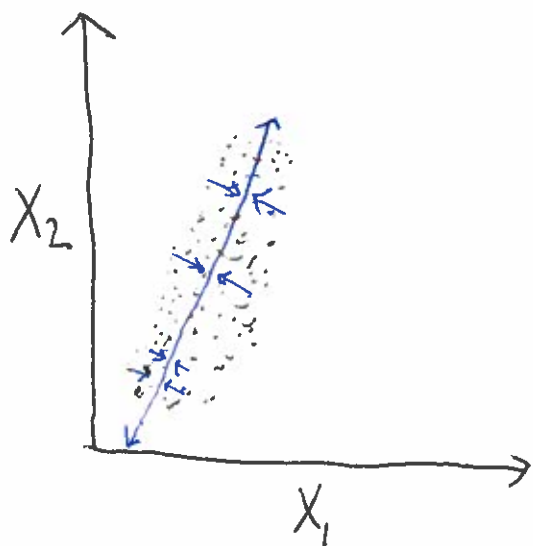
Here, I'm going to take just a few moments to go through the same high-level visual/intuitive explanation of principal components analysis (PCA) that is typically used to first introduce the topic. This should mostly serve as a refresher to give us some common intuition and bring it to the front of our minds.

We will go over a very simple two-dimensional example:



We want to find the directions that maximize the variance of this data. In particular, we want to choose the direction/vector where if we projected our data onto it and then computed its variance, which direction would be maximal? To see the idea of a "projection" better, let's look at the plot on the right, which shows the range of the data when projected onto x_1 & x_2 (which is equivalent to just taking the value x_1 or x_2 , respectively). Note that since there is more spread in the x_2 feature, the projection onto x_2 captures more of the data's variance than the projection onto x_1 .

However, of course, we can do better.



By choosing the diagonal line that captures most of the spread in the data, we can maximize the variance of the projected data.

For the second principal component, if we wanted to also maximize variance (and were not restricted in any way), there are an infinite number of options: we would want to choose a line/diagonal just like the first one, but maybe tilted a tiny amount, since that projection would also have very high variance.

However, we have one constraint with PCA: the directions have to be orthogonal (should be ringing a bell in your head about the U matrix in SVD). So in the 2D case, there is only one choice, the line perpendicular to the first principal component.